

SOFTWARE–HARDWARE SYSTEMS

FUNDAMENTALS OF THE INTEGRATED USE
OF NEURAL NETWORK AND ONTOLOGICAL
PARADIGMS: A COMPREHENSIVE APPROACH*O. Palagin,^{1†} V. Kaverinskiy,² K. Malakhov,^{1‡} and M. Petrenko^{1††}

UDC 004.89

Abstract. *The paper presents an integrated approach that combines neural network and ontological paradigms. The method encompasses methodological underpinnings, information technology, and the MedRehabBot system. Collectively, they embody the core principles of meta-learning and structured prompts, ultimately enhancing the efficiency of information system interaction with Chatbots and information retrieval rooted in ontologies. The method also offers the flexibility to adapt the MedRehabBot system for utilization within different Large Language Model (LLM) systems.*

Keywords: *transdisciplinary scientific research, ontology, ontological engineering, ontology-driven information system, LLM-system, ChatGPT, MedRehabBot, prompt engineering, Chatbot.*

INTRODUCTION

The present stage of development of science and its applications has a distinctly transdisciplinary (TD) nature. “This stipulated the necessity of the development of a rigorous methodology of transdisciplinary (TD) scientific investigations, the creation of TD international centers and schools, and, finally, the determination of the place and role of informatics in the system-technological support of TD investigations and use of their results in solving global problems of developing modern civilization. The TD paradigm presumes the construction of a general scientific picture of the world in the foreseeable future or, what is the same, a unified TD knowledge system providing formalized statements and solutions to concrete problems in executing integrated projects of high complexity, social significance, proneness to conflict, and competitiveness” [1]. In other words, it is about metascience capable of explaining the processes and phenomena of the surrounding world and solving applied problems of nature and society, which are becoming increasingly complex.

The TD research encompasses zones of boundary (demarcation) areas of scientific disciplines, integrating the essential foundations of the latter, forming clusters of convergence where powerful synergistic interaction occurs through the interpenetration of paradigms and specific current results of each discipline into one or another cluster. This interaction reflects the integrity of the real world. It serves as a stimulus (and a guarantee of success) for conducting TD research and implementing associated practical projects and points to the non-triviality and significance of their results [1, 2].

*The study was supported by the National Research Foundation of Ukraine (Grant # 2021.01/0136 “Development of the cloud-based platform for patient-centered telerehabilitation of oncology patients with mathematical-related modeling”) and conducted at the V. M. Glushkov Institute of Cybernetics, National Academy of Sciences of Ukraine, Kyiv, Ukraine.

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A crucial step towards transdisciplinarity is the formation of prospective self-sufficient clusters of transdisciplinary research that take into account the consequences (risks) and mutual influence of core factors and feedback in the process of theoretical analysis, purposeful physical experiments, and the implementation of global systemic projects for sustainable development of human society, environmental conservation, overall scientific development, etc. [3]. A vivid example could be the cluster of NBIC (Nano-Bio-Info-Cogno) convergence, which has defined breakthrough directions in the development of cutting-edge technologies blurring the boundaries between living and non-living systems, nanorobotics with its numerous applications, global supercomputer agglomerations with high levels of artificial intelligence, etc. Another example is the unified distributed TD knowledge system, which serves as a global communicative version of a general scientific worldview and signifies the next stage of development beyond the existing Internet and the Semantic Web networks.

It is believed that TD research should ensure effective TD interaction at all stages of the life cycle in solving fundamental and applied scientific problems; methodological support and facilitation of integration, convergence, and unified formalized presentation of TD knowledge, as well as operations upon them; a shared basis for mutual understanding among representatives of different scientific disciplines; the inclusion of humans in the intra-scientific context, the overall convergence of natural and humanitarian knowledge, etc.

Given the transdisciplinary nature of modern scientific programs and projects, methods for analyzing the essential processes of convergence between subject areas and the associated processes of cognitive interaction at the level of conceptual structures appear to be particularly relevant. The above directly relates to the information system MedRehabBot, which can become a highly productive assistant, specifically in developing new clusters of convergence between subject areas using artificial intelligence (AI) tools.

The use of natural language in human-computer interaction has always been relevant, most notably due to its convenience compared to learning specialized languages or instructions. Instead of learning complex interfaces, users could express their intentions in natural language. Implementing natural language into interactive systems will enhance their intuitiveness and effectiveness. Virtual assistants like Apple Siri, Amazon Alexa, and Google Assistant leverage these capabilities already. The ChatGPT system of dialog services from OpenAI relies on GPT (Generative Pre-trained Transformer) LLM (Large Language Model) models such as GPT-3, GPT-3.5, GPT-3.5-turbo, and GPT-4 [4, 5], refined for natural language applications using supervised and reinforcement learning methods [4, 6]. Its emergence marked a significant breakthrough in AI, particularly in natural language processing (NLP) and natural language understanding (NLU), and stands as a substantial step forward.

We have carried out our research to continue, develop, and improve upon the previous work [7]. It enabled the development and testing of a state-of-the-art interactive information system named MedRehabBot to support physicians in physical and (tele-) rehabilitation medicine (PRM), PRM specialty students, and patients. It allows for providing information and performing logical inferences based on specific contextual sets, functioning as a dialog system. We carried out this work as part of the project titled “Development of the cloud-based platform for patient-centered telerehabilitation of oncology patients with mathematical-related modeling,” supported by a grant from the National Research Fund of Ukraine [8–10]. The authors are members of the Interagency Working Group on Developing the Concept of Telemedicine Implementation in Ukraine, specifically the Subgroup on Technical Issues and Architecture of Telemedicine.

We have demonstrated the implementation of the method for improving the efficiency of interaction between ontolinguistic and neural network tools using a set of scientific papers written in Ukrainian and English in the field of physical and PRM, particularly e-rehabilitation [11].

The OpenAI platform [12] has a limited Application Program Interface (API), characterized by the capability to send target prompts in natural language without a clearly defined structure. For more convenient use of the API and to specify parameters and settings, the LangChain wrapper package has been created [13–15]. The primary language for OpenAI’s GPT models and ChatGPT is English [4]. Due to the limit on the number of tokens that ChatGPT can process in a single query, instructions should be brief but informative. Experimental data on using ChatGPT [7, 16, 17] showed that one efficient approach for providing concise yet comprehensive commands and instructions is their representation in JSON format. This method ensures the ordering, ease, interpretability, and maximum efficiency of information for ChatGPT. Such an approach allows for clearly defining desired actions and expectations, optimizing interaction between users and ChatGPT [18–20].

Papers [21, 22] discuss using ontologies as repositories for behavior rules, albeit without direct references to ChatGPT. Ontologies can contribute to decision-making and data representation in an interface. Paper [23] describes an ontology-driven architecture.

MODELS OF INTERACTIVE REFERENCE-INFORMATION SYSTEM “MedRehabBot” TO SUPPORT REHABILITATORS, STUDENTS, AND PATIENTS IN THE FIELD OF PHYSICAL REHABILITATION AND TELEREHABILITATION

The information model and development toolkit of the MedRehabBot system. The MedRehabBot system utilizes an information model based on a composite service (S) from a three-component tuple [7, 24, 25]. This model enables the integration of various services within the following system:

$$S = \langle D, F, E \rangle,$$

where $D = \{ws_w, as_d \mid w = \overline{1}, \overline{k}, d = \overline{1}, \overline{l}\}_{k, l = \overline{1}, \overline{N}}$ is a set of web services (ws_w) and application software (as_d) available for developers, enabling the development of various applications and services within the system, i.e., the set for developing MedRehabBot, N is the set of positive integers; $F = D : \{C_j \mid j = \overline{1}, \overline{n}\}_{n = \overline{1}, \overline{N}}$ is a set of functions covering the operational aspects of the information technology implemented by the MedRehabBot system. Each function corresponds to the specific conveyor or knowledge control process resulting from the integration and interaction of elements within D ; $C_j \subseteq D$, $C_j = \{ws_o, as_p \mid o, p \geq 0, o \leq k, p \leq l\}_{o, p = \overline{1}, \overline{N}}$ is a subset of web services and application software required for the successful implementation of the j th function within D ; $E = \{prl, os, floss\}$ is a set of elements that unite within a layered structure and form an integrated knowledge development environment (K-IDE) [24].

Element *prl* represents the physical equipment and resources of the object, as defined in [24]. The element *os* belongs to the operating system (OS) level and represents a guest OS in K-IDE. The OS level is intended for using Unix-like OS, such as Ubuntu Server for x86 systems, and DietPi, a lightweight Debian-based OS for single-board computers (SBC) based on ARM architecture. This level provides the basis for K-IDE framework operation and compatibility with selected OSes and desktop computer environments. The FLOSS level, denoted as *floss*, represents the component of free and open-source software (FLOSS) within K-IDE. This level covers both internal and external software components. Paper [7] extensively describes the formal representation of web services *ws* and *as* and their constituents.

Functional model of the MedRehabBot system. The functional extension of the MedRehabBot system is represented by the following set of F functions synthesized from D :

$$F = \{C_1, C_2, C_3\},$$

where C_1 is the automatic generation of an OWL ontology (multiset of RDF triples) formally describing natural language text; C_2 is the function of ontology-driven dialogue, integrating ChatGPT and structured prompts; C_3 are structured prompts for the meta-learning function of ChatGPT.

METHODOLOGICAL PRINCIPLES OF CONSTRUCTING AND USING THE MedRehabBot SYSTEM

We can divide the methodology into two main parts, each serving a separate goal in developing the MedRehabBot system.

The first part focuses on the meta-learning technique based on creating structured prompts for ChatGPT. This approach controls the meta-learning process of ChatGPT, allowing the generation of more contextually relevant and precise responses. The procedures for developing and implementing these prompts are detailed, emphasizing their importance in enhancing the conversational abilities of ChatGPT.

The second part focuses on developing an automatic ontology-driven dialogue system integrating ChatGPT with structured prompts. The key idea of this system is the efficient manipulation of contexts within specific subject domains, which may contain domain-specific information not entirely covered by ChatGPT's knowledge base. One stores these contexts in a database, e.g., a document-oriented database management system (DBMS) like MongoDB or an object-relational DBMS like PostgreSQL. Contexts are associated with sets of named entities having their own structure similar to an ontology. Additionally, we can employ a narrative analysis for categorizing these contexts. Linking named entities to their respective contexts involves semantic components explaining the entity's role in context. These additional features enhance the relevance and clarity of the selected context for further processing. A previously developed toolkit is used for process automation [2, 7, 26].

For semantic analysis and extraction of named entities from user-provided phrases, ChatGPT proves to be a valuable resource. Specialized prompts have been created specifically for this purpose. Moreover, ChatGPT is used to analyze the intents behind user phrases. We present identified intents, extracted named entities annotated with their semantic roles, and a selected list of contexts as input to ChatGPT. These input data are accompanied by corresponding structured prompts that clarify the information to be extracted and the desired format for its presentation.

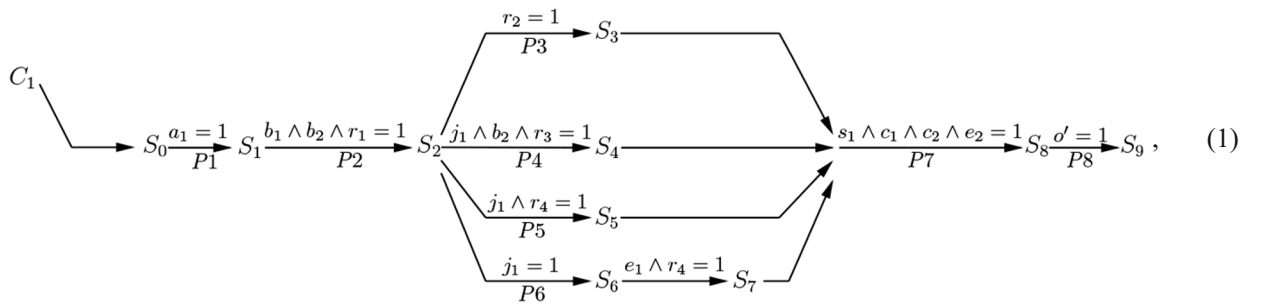
Paper [7] presents the general diagram of the C4 model [27] (specifically, the context and container).

One of the main system's features is the flexibility of structured prompts for ChatGPT. These are dynamically generated instead of fixed prompts, depending on the specific situation, using instructions provided in the form of metaontology. The metaontology describes the fields we should include in the JSON (or XML) format structure and the corresponding prompt phrases we should insert. Each instruction or structured prompt for ChatGPT has its own set of fields and predefined values that we can include. Additionally, the prompt contains a template structure for the response, ensuring consistency and simplifying further processing. Verified methodologies from [18, 28, 29] are employed in constructing prompt phrases to achieve effective and coherent prompts.

The structure of the MedRehabBot system includes several components, states, procedures, and variables that contribute to the functionality and operation of the dialogue system. Below is the overview of its key elements.

We can depict the operation of the MedRehabBot system using a scheme similar to the Petri net [30], specifically in the form of the modified marking graph of system network [31]. This representation ensures a structured depiction of the system's functioning, depicting the flow and interaction among various components. We illustrate a more detailed examination of the functioning of the graph components by expressions provided below.

The automatic creation of an ontology for contexts can employ various approaches depending on the nature of source data and the system's operational specifics. Below is one possible example of automatically creating the ontology based on a set of source files in PDF format with regular, known beforehand structure. In this case, the basic structure of the ontology is also known. We can describe the abovementioned component C_1 using the following modified Petri net:



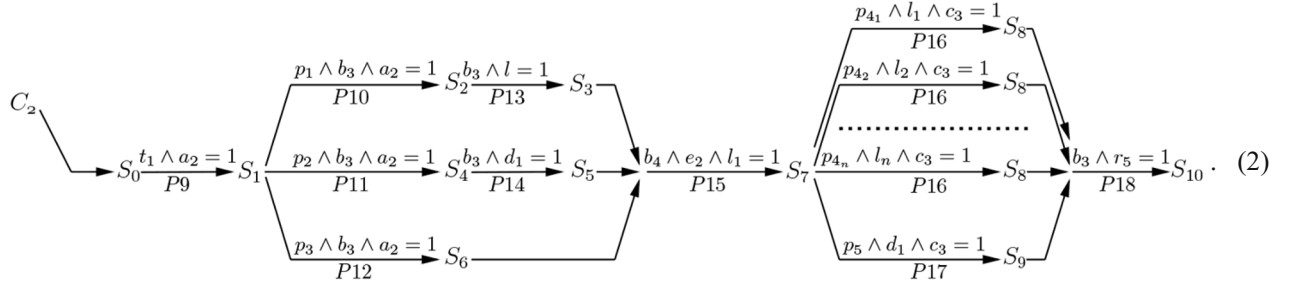
where states for component C_1 (from S_0 to S_9) are as follows: S_0 is the initial state (a set of PDF files); S_1 is the list of textual representations of PDF documents; S_2 is the set of documents in structured JSON representation; S_3 is the program representation of basic ontology components independent of source data. It includes the description of declarations and predicates and top-level classes and properties, which are basic elements. In this case, for instance, these properties encompass those responsible for linking contexts to specific articles (documents) and named entities to respective contexts; S_4 is the program representation of ontology classes; S_5 is the program representation of ontology instances (Named Individuals); S_6 are named entities extracted from respective contexts; S_7 is the program

representation of named entities as instances (Named Individuals) in the ontology; S_8 is the integrated program representation of the created ontology; S_9 is the file containing the OWL description of the context ontology.

The procedures in expression (1) are as follows: $P1$ is the extraction of textual data from source PDF files. Text is obtained for each file as a list of strings; $P2$ is the derivation of JSON structures from the textual data in a specified format, filled in with corresponding contexts; $P3$ is the construction of program representation of the basic ontological structure in the specified format; $P4$ is the creation of program representation for a set of ontology classes. Classes are primarily generated based on the keys of JSON dictionaries for document representations, considering the specified structure template; $P5$ is the generation of program representation for a set of ontology instances (Named Individuals) based on contexts or named entities; $P6$ is the recognition of named entities from contexts (named entity recognition); $P7$ is the compilation of program representation of the ontological structure from obtained classes, properties, and instances; $P8$ is the serialization of the representation of the created ontological structure into the OWL file.

Expression (1) contains the following variables: a_1 is the initial set of PDF documents; b_1 is the list of textual representations of PDF documents; b_2 is the specified structure for the created JSON representations; r_1 is the set of rules and instructions for parsing textual representations of documents; r_2 is the set of rules and instructions for building the overall ontology structure; j_1 is the collection of JSON representations of source documents; r_3 is the set of rules and instructions for building program representations of ontology classes; r_4 is the set of rules and instructions for building program representations of ontology instances; e_1 is the collection of named entities extracted from contexts; s_1 is the program representation of the overall ontology structure; c_1 is the program representation of ontology classes; c_2 is the program representation of ontology context instances; e_2 is the program representation of named entity instances in the ontology; o' is the program representation of the ontology of contexts.

Component C_2 in the expression below is an ontology-driven dialogue function that integrates ChatGPT and structured prompts:



Expression (2) illustrates the dialogue process between the user and the MedRehabBot system, where the states for component C_2 (from S_0 to S_{10}) are as follows: S_0 is the initial state of receiving a text message from the user; S_1 is the state of preprocessed text prepared for further operations; S_2 is the state of the list of identified intents; S_3 is the state of the formed template of a request for extracting information; S_4 is the state of logical inference that needs to be executed; S_5 is the state of the formed template of a prompt for generating conclusions; S_6 is the state of the extracted list of named entities; S_7 is the state of the list of selected contexts; S_8 is the state of the information extracted from contexts; S_9 is the state of conclusions obtained from contexts; S_{10} is the state of the results obtained in a user-friendly format.

The procedures in expression (2) are as follows: P_9 is the initial text processing; P_{10} is determining intents expressed in the input text; P_{11} is determining if the intent implies generating conclusions, prompting ChatGPT to provide relevant information from the given contexts or associated knowledge; P_{12} is the recognition of named entities from the input text; P_{13} is the formulation of prompts for determining intents; P_{14} is the formation of a prompt for generating conclusions; P_{15} is the selection of relevant contexts based on identified named entities and intents; P_{16} is extracting information from selected contexts according to the intent; P_{17} is the generation of conclusions from selected contexts; P_{18} is formatting and presenting results to the user.

Variables in expression (2) are as follows: t_1 is the set of rules and operations that will be applied during the pre-processing of the input text; a_2 is the initial text provided by the user; a_3 is the preprocessed text prepared for further operations; b_3 is the metaontology containing operational rules and instructions for composing operational messages; b_4 is the ontology of contexts associated with the system; p_1 is the prompt for intent determination; p_2 is the prompt for analyzing whether a particular action needs to be taken; p_3 is the prompt for extracting a named entity; $p_4, \dots, p_N$ are prompts for

extracting information based on a specific intent $n, n = \overline{1, N}$; p_5 is the prompt for ChatGPT to generate conclusions from specified contexts; l is the set of intents determined in the input text triggered according to specific entities; l_n is the specific unique intent, $n = \overline{1, N}$; d_1 indicates whether conclusions will be generated, and if so, what specific conclusions are expected; e_2 is the set of named entities found in the input text; c_3 are the contexts selected; r_5 are the structures of data representing the obtained results from ChatGPT.

We can describe the procedure of a dialogue act as follows: upon receiving from the user, textual information (a query) undergoes intent analysis using ChatGPT with the help of a specialized prompt. The result is a list of dictionaries containing the following keys:

- “name”: the name of an intent from the provided query list;
- “type”: a more general classification of the intent, such as “narrative,” “interrogative,” or “imperative;”
- “probability”: a floating-point value ranging from 0 to 1 indicating the likelihood of the intent’s presence in the user’s text;
- “subject”: the subject associated with the intent, if such an association is possible;
- “object”: the object associated with the intent, if such an association is possible.

The query may contain various possible intents such as “quantity,” “way of doing,” “object,” “subject,” “action,” “location,” “direction,” “scene of action,” “conditions,” “instrument,” “collaborator,” “relation,” “cause,” “sequence,” “origin,” etc. These intents represent semantic categories and provide the basis for understanding the user’s query. Moreover, the structured query includes fields for providing information, language, and other technical details related to the input and output of data.

Simultaneously, using ChatGPT and another prompt, named entities are extracted. The result should contain lemmatized words grouped by entities, the type of each group (noun or verb), and the main word within each group.

Based on the named entities extracted from the intents and their semantic roles, the system selects relevant contexts from the ontology of contexts.

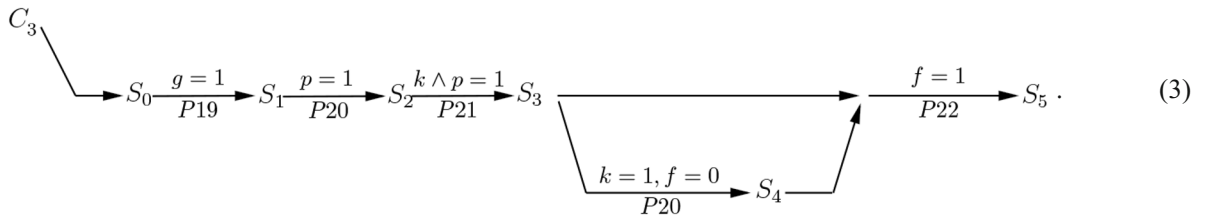
Next, one makes a query to ChatGPT, which includes the previously selected contexts, lists of intents, and corresponding entities (subjects and objects) updated in the prompts. The form of this query can significantly vary and depends on a particular intent or intent list. The rules of constructing the list, including mandatory fields and prompt phrases, are defined in the metaontology.

The metaontology also encompasses rules for the final presentation of results, depending on the types of fields used. The format for representing the results can include plain text, numbers, dates, lists, tables, or links to external resources. There are other potential ways to present the results as well.

Creating metaontology involves meta-learning, which encompasses developing and refining relevant prompts. In this iterative process, a knowledge engineer (as a distinct actor) and the playground of the OpenAI platform participate. Initially, the knowledge engineer formulates a structured prompt in JSON (XML, YAML, etc.) format, consisting of keys and prompt phrases designed for a specific purpose, drawing from previous prompt development experience and general knowledge. Then, one provides this prompt to ChatGPT and analyzes the response to determine if it aligns with the intended goal.

If the response from ChatGPT is entirely incorrect or contains shortcomings, one changes the initial prompt. These changes may involve adding additional fields, removing unnecessary or ineffective aspects of the prompt, and editing prompt phrases to enhance the clarity of the task. Then, a user resubmits the improved prompt to ChatGPT for further iterations. This iterative process continues until the obtained response fully aligns with the desired and expected result.

After developing a relevant set of prompts and studying ChatGPT’s behavior with these prompts, they are consolidated into a metaontology format. Creating a metaontology is a manual process where the prompt structure and properties of relevant fields are described based on various anticipated situations and task-solving objectives. We present the formalized graphical representation of component C_3 in the meta-learning process by the following expression:



Here, the states for component C_3 (from S_0 to S_5) are as follows: S_0 is the purpose of creating a prompt; S_1 is the initial prompt created based on guesses and wisdom of life; S_2 is the ChatGPT's response to the initial prompt; S_3 is the revised and edited prompt; S_4 is the ChatGPT's response to the revised prompt; S_5 is the metaontology with instructions and rules for the new prompt.

The procedures in expression (3) are as follows: $P19$ is the expressing informed guesses and considerations regarding the new prompt; $P20$ is the initializing ChatGPT; $P21$ is the revising and editing the prompt; $P22$ is the integrating the new prompt into metaontology.

The variables used in this expression are as follows: g is the initial idea for creating a new prompt; p is the initial prompt; p' is the refined query version; f is the final query version; k is the ChatGPT's response to the initial query; k' is the ChatGPT's response to the modified query.

RESULTS OBTAINED

We have developed a prototype of the proposed MedRehabBot system that includes all the main components. Detailed working materials can be found in the public GitHub repository [32] associated with this paper. The prompts in the system are formulated in the form of JSON structures. Below is the JSON schema representing an example prompt aimed at determining intents and their relationship with relevant entities (if such a determining is possible):

```
{
  "information to provide": [
    "define intents",
    "find subjects",
    "find objects"
  ],
  "text": "<A text to be analyzed>",
  "language": "Ukrainian",
  "input information field": "text",
  "possible intents": [
    "quantity", "place", "way of doing", "object", "subject", "action", "location",
    "direction", "scene of action", "conditions", "instrument", "collaborator",
    "relation", "cause", "sequence", "origin"
  ],
  "several intents": true,
  "intents probability": true,
  "show intent subject": true,
  "max intents number": 4,
  "intents arrange": "by probability",
  "output format": "JSON",
  "output representation template": {
    "result": [
      {
        "intent": "intent name - string",
        "type": "narration, interrogation or imperative",
        "probability": "float value",
        "subject": "subject of the intent as a name group – string",
        "object": "object of the intent as a name or verb group – string"
      }
    ]
  }
}
```

TABLE 1. Test Results of the Proposed Method

Response	True	False
Positive	27	2
Negative	6	3

An essential part is the “information to provide,” which formulates the main goals of this task prompt. The text of the input message subject to analysis is indicated in the “text” field. To increase productivity, we have specified the input text language. In this example, it is Ukrainian. The “possible intents” list is used to precisely establish the set of intents that can be identified in the input text. Several boolean-type fields are designated to provide technical information about the output. In this case, they are presented as “several intents,” “intents probability,” and “show intent subject.”

The first field allows for identifying several intents in the provided text. The second field helps ChatGPT assess the probability of each intent. The last one indicates entities specifying the intents. To avoid excessive intent recognition, we can limit intent number (we have selected four intents in this study) and order them by probability. For simplifying the subsequent processing of results, it is recommended to define a template for the structure of output data, specifying the format in which information is returned. This action is controlled by the “output representation template” field.

We have tested the proposed method using the MedRehabBot system, built on the information sources in physical and rehabilitation medicine. To create the knowledge base, we used the “EBSCO articles dataset (domain knowledge: rehabilitation medicine) + JSON of every article” [33] dataset and constructed an ontology of contexts based on it [32]. We have performed the classification of the obtained responses as follows:

- “True Positives (TP)”: the response was generated by ChatGPT and turned out to be correct;
- “True Negatives (TN)”: ChatGPT acknowledged its lack of knowledge or indicated insufficient information. This category also includes cases where ChatGPT could not provide a response, and the relevant information was absent in the contexts.
- “False Positives (FP)”: the system attempted to provide a response, but it turned out to be incorrect;
- “False Negatives (FN)”: ChatGPT did not provide a response even though the correct answer was available in the contexts.

During the testing phase, we excluded questions and phrases from unrelated subject areas containing names of entities not present in the contextual ontology. If we do not do this, we will obtain a negative result because no relevant context is selected, and further processing will not occur. Thus, we formulated all tests to align with the proposed approach.

Note that the proposed approach allows for several answers to a single question, primarily due to the presence of multiple defined intents. During evaluation, we considered all provided responses, regardless of whether they responded to the same question or different ones.

Manipulating experimental data allows for obtaining a comprehensive assessment of the system’s performance and evaluating its ability to process various queries, considering defined intents, and extracting relevant information from selected contexts.

Table 1 presents the test results.

The testing of the proposed method yielded the following values for standard evaluation metrics:

- Accuracy: 0.8684;
- Precision: 0.9310;
- Recall: 0.9000;
- F1 Score: 0.9152.

These metrics provide a quantitative assessment of the system’s performance in terms of its accuracy, precision, recall, and overall effectiveness. The accuracy metric reflects the proportion of correct responses provided by the system compared to the total number of queries. It measures the system’s ability to provide accurate responses among those it generates. Precision indicates the system’s capability to retrieve all relevant responses from available contexts. The F1 score combines precision and recall to provide a balanced assessment of overall performance.

The obtained metric values demonstrate the potential suitability of the proposed method, although we cannot yet consider it perfect.

We addressed a drawback identified in our previous study [7], which involved a relatively low accuracy due to the high probability of false-positive responses. This improvement stems from a more refined structure of the ontology of

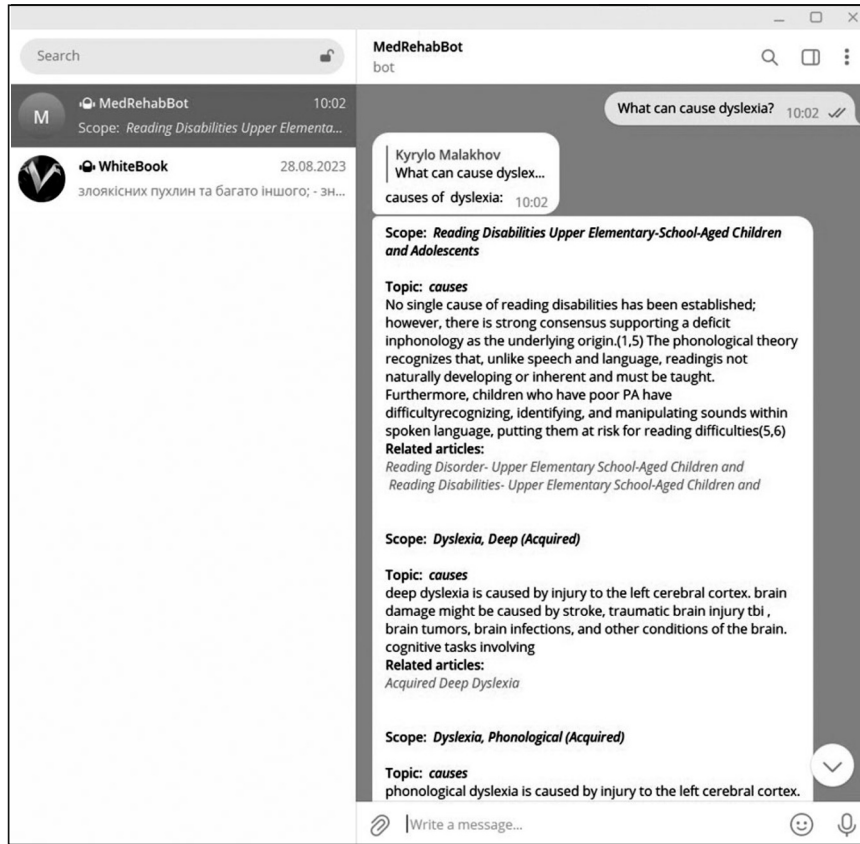


Fig. 1. Graphic user interface of the software implementation of the MedRehabBot system (in the form of an application for the Telegram messaging system).

contexts and queries to it, isolating more relevant contexts that do contain the answers. Consequently, ChatGPT no longer needs to search for approximate information in the extensive but not always relevant context it receives.

Also worth noting is that within the scope of the considered problem, false-positive responses often accompanied true-positive ones and didn't inherently pose a significant problem. Thus, we could regard them as additional information relevant to the primary response. This observation even underscores the potential value of considering false-positive responses in a practical context.

Let us consider the following example. Figure 1 shows the graphical user interface of the MedRehabBot system's software implementation (in the form of an application for the Telegram messaging system). The initial query is: "What can cause dyslexia?" The system recognizes the following intents:

```
[
  {
    "intent": "cause",
    "type": "interrogation",
    "probability": 0.8,
    "subject": "",
    "object": "dyslexia"
  },
  {
    "intent": "origin",
    "type": "interrogation",
    "probability": 0.7,
    "subject": "dyslexia",
```

```

    "object": ""
  },
  {
    "intent": "way of doing",
    "type": "interrogation",
    "probability": 0.6,
    "subject": "cause dyslexia",
    "object": ""
  },
  {
    "intent": "action",
    "type": "interrogation",
    "probability": 0.5,
    "subject": "can cause dyslexia",
    "object": ""
  }
]

```

To select contexts, the following named entities have been found:

```

[
  {
    "words": ["cause", "dyslexia"],
    "type": "noun",
    "main word": "cause"
  },
  {
    "words": ["dyslexia"],
    "type": "noun",
    "main word": "dyslexia"
  }
]

```

We have determined the following intents: “cause”, “origin”, “way of doing”, and “action”. In all cases, the sentence is determined as “interrogation.” All these intents have relatively high probabilities, and we should consider them to obtain the final response set. The responses obtained are as follows:

— “No single cause of reading disabilities has been established; however, there is strong consensus supporting a deficit in phonology as the underlying origin. The phonological theory recognizes that, unlike speech and language, reading is not naturally developing or inherent and must be taught. Furthermore, children who have poor PA have difficulty recognizing, identifying, and manipulating sounds within spoken language, putting them at risk for reading difficulties” (“cause”).

— “Deep dyslexia is caused by injury to the left cerebral cortex. Brain damage might be caused by stroke, traumatic brain injury TBI, brain tumors, brain infections, and other conditions of the brain cognitive tasks involving” (“origin,” “way of doing,” “action”).

The system provided two response options. The second option, obtained based on the intents (“origin,” “way of doing,” and “action”), is more direct and points to the physiological causes of severe dyslexia. We can consider that the first option is also relevant and complements the second option in content.

Generally, the reference system may be limited to providing only a set of ready-made text fragments from the context ontology. However, processing them through a powerful language model (such as ChatGPT-3.5-turbo) with a specific intent set allows for generating the more definite and clear response to the posed question, where contexts serve as a source of information.

As mentioned above, the proposed approach provides ChatGPT with additional information and expands its capabilities to provide responses compared to using only the available knowledge base. Below is an example illustrating such a comparison between the performance of just ChatGPT and ChatGPT with the provision of a prompt containing intents and contexts. The text of the question is as follows: “What is the ICD-10 code for breast cancer?”

ChatGPT provides the following response: “The ICD-10 code for breast cancer is C50.” It is correct. The proposed model using context transfer from the pre-built ontology also provides the value of code C50. However, it additionally offers an explanation: “a personal history of malignant neoplasm of the breast (ICD codes are provided for the reader’s reference, not for billing purposes).” Moreover, this model provides a set of links to relevant articles and broader information about ICD-10 codes associated with the mentioned subject area, such as Z90.1 — acquired absence of the breast; I97.2 — postmastectomy lymphoedema syndrome; Z85.3 — personal history of malignant neoplasm of the breast; Z80.3 — family history of malignant neoplasm of the breast; R92 — abnormal findings on diagnostic imaging of the breast.

Let us consider another example based on the query of the ICD-9 code: “What is the ICD-9 code for breast cancer?”. There is no ICD-9 code for this case in the contextual ontology. Therefore, we obtained the standard response as follows: “Sorry, but I don’t know about the ICD-9 code for breast cancer.” In turn, ChatGPT returns the following message: “The ICD-9 code for breast cancer is 174.9. Please note that ICD-9 codes have been largely replaced by ICD-10 codes for medical coding and classification.” Hence, the peculiarity of the proposed approach is that ChatGPT formulates responses based namely on the transmitted contexts and the information presented (or absent) within them.

CONCLUSIONS

As the result of our research, we have developed a productive triad encompassing three key components: the methodological principles of using ontology-driven structured prompts in ChatGPT meta-training, new information technology, and the comprehensive MedRehabBot service. The latter is an advanced interactive information support system for physical and (tele) rehabilitation medicine practitioners, students specializing in PRM, and patients. We have created this system to advance and enhance our previous research [7].

We have developed a method to enhance the efficiency of interaction between ontolinguistic and neural network tools, involving meta-training for creating and configuring structured prompts and contextual ontology. It includes a sequence of operations for determining intents, recognizing named entities, selecting contexts, and forming the final prompt and response based on information and conclusions provided within contexts. The key feature of this method is its ability to provide ChatGPT with specific information using selected contexts, expanding its knowledge in concrete subject domains and improving its performance when working with data in different languages.

Note that we can apply the proposed methodological principles not only to ChatGPT and OpenAI’s GPT models used in the research but also to the other chatbot systems based on LLM models (such as Google Bard). One can adapt the fundamental principles and methods of meta-training, structured prompt formation, and ontology-based information retrieval and use them combined with various LLM systems. This potential universality and scalability underscore the broader applicability of the proposed approach across different chatbot platforms, enabling its wider use in natural language processing and conversational systems.

The ontology in the MedRehabBot information system provides the following effects:

- Structured knowledge;
- Semantic understanding;
- Increased precision;
- Expansion and adaptability of knowledge to specific domains;
- Interoperability;
- Knowledge-driven prompts;
- Support for meta-training.

More generally, the proposed approach can be a prototype for developing a more sophisticated user-system dialogue. We plan several improvements for the structured prompts in JSON format to enhance system performance. We will add to the prompts supplementary input data, specifically the intents identified in the initial message and selected contexts. We anticipate that these enhancements will improve the accuracy criterion of the system.

Future research involves integrating the proposed comprehensive method for enhancing the interaction between ontolinguistic and neural network tools with medical information systems of rehabilitation type and hardware support in dialog systems using logical hardware technologies for its implementation [34].

The software implementation of the MedRehabBot system (in the form of a messaging application for Telegram), the metaontology, context ontology, SPARQL queries to the metaontology, samples of structured JSON prompts for ChatGPT, test queries, and results are available in the open-access repository on GitHub at the following link: <https://github.com/knowledge-ukraine/MedRehabBot>.

The dataset “EBSCO articles dataset (domain knowledge: rehabilitation medicine) + JSON of every article” is available on the Zenodo platform at the link [33].

The demo version of the MedRehabBot software implementation is available for exploration and evaluation as an application for the Telegram messaging system: <https://t.me/MedicalRehabBot>.

You can obtain detailed information about the usage and access to the MedRehabBot system by contacting the team of authors.

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